

# WHICH BIAS FLAGS CAN YOU TRUST? A LABEL-FREE TRIAGE PROTOCOL FOR FRONTIER-MODEL NEWS-BIAS DETECTION

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## ABSTRACT

News-bias products—bias checkers, balanced-news feeds, media-literacy tools—are increasingly built by asking a frontier model to flag biased spans, label an article’s lean, and explain itself. Which of those outputs can a builder trust, and how should the untrustworthy ones be triaged *without* ground-truth labels at inference time? Evaluating two frontier target models and two cross-family judges on 200 expert-rated U.S. news articles, we first map the primitives: the *volume* of flagged spans is a valid partisan-intensity meter (Spearman  $\rho \approx 0.6$ ), and models rank-track the continuous expert lean rating at  $\rho \approx 0.8$ , but a single model’s lean label near the center-right, its omission/framing flags (chance-corrected agreement  $\kappa \leq 0.14$ ), and the specific text it highlights (cross-model overlap 0.26) are model-specific opinion. We then turn the map into a validated selective-prediction protocol. A cost-free *committee-disagreement* bit carries the same error-ranking information as a calibrated confidence score (AUC 0.69 vs 0.70,  $p=.90$ ) for free, captures significantly more errors per unit of human-review budget in the practical 20–30% regime (+6–8pts,  $p<.05$ ; disagreements are 32% vs 71% accurate), and combined with confidence dominates either signal alone on the risk–coverage curve; a 4-model unanimity gate clears 44% of articles at 87% accuracy; and neither raw confidence (both models overconfident; a stated 0.9 means  $\sim 75\%$ ) nor a majority-vote ensemble (no accuracy gain over the best single model) should be used as a probability. Every result holds across five prompt conditions, and the reliability map and the directional failure both replicate on a six-model committee spanning four independent labs (Anthropic, OpenAI, Alibaba, DeepSeek). Findings are exploratory and inherit AllSides outlet labels as ground truth; all statistics carry bootstrap CIs.

## 1 INTRODUCTION

A growing class of products puts a frontier language model in the role of a news-bias analyst: it highlights loaded spans, assigns a left/right lean, summarizes “both sides,” and narrates why. The implicit assumption is that these outputs are measurements. Many are not—they are one model’s opinion, and a different frontier model, shown the same article, would say something else. A builder needs two things before shipping: to know *which* outputs reproduce, and—for the ones that do not—a way to catch the likely errors that needs *no* ground-truth labels at inference time, since a deployed product has none.

This paper supplies both. We do not ask whether the models are biased (a companion study takes up the behavioral question); we ask which of a bias detector’s outputs a tool can present as reliable, and how to triage the rest. Our answer is a reliability map with a clean split—primitives anchored to explicit text reproduce across independent models; primitives that require inference about what is absent or implied do not—and, built on it, a label-free triage protocol whose central signal is not model confidence but *model disagreement*. We validate the protocol against the standard confidence baseline with a risk–coverage operating curve, and show it is stable across prompt conditions and across four independent model vendors rather than an artifact of one.

**Contributions.**

1. **A reliability map of bias-detection primitives.** Which outputs of a frontier-model bias detector reproduce across models (detection volume; continuous-rating rank; cross-model agreement) and which are model-specific opinion (lean labels near center-right; omission and framing flags; specific highlights)—each with a number and a bootstrap CI (§4).
2. **A validated, label-free triage protocol.** A cost-free committee-disagreement router with a published risk–coverage operating curve, benchmarked against a calibrated confidence router (equal global ranking, superior in-budget error capture); a high-precision unanimity gate; and a per-model calibration analysis showing raw confidence must be rescaled and is trustworthy only for direction (§5).
3. **A directional failure mode shared across four vendors.** All six models we evaluate—across four independent labs—read moderate-right content as center-or-left, so a lean-labeling tool built on them under-counts the right; stable across five prompt conditions (§6).

## 2 RELATED WORK

**LLM-as-judge reliability.** Using an LLM to score model outputs is now standard (Zheng et al., 2023), and its biases—position, verbosity, self-preference—are documented (Ye et al., 2025; Panickssery et al., 2024). Reliability with consequences is emerging at the *instance* level (selective evaluation with human-agreement guarantees and escalation (Jung et al., 2025)) and judge *selection* by human agreement is practiced in benchmark construction (Dubois et al., 2024; Lambert et al., 2024). We differ in object and use: rather than validating a judge for scoring, we ask which *primitives* of a frontier bias detector a downstream product can trust, and we supply a deployment-time triage signal that needs no human labels at all.

**Selective prediction.** Deferring uncertain cases to a human is classical selective classification, evaluated by the risk–coverage curve (El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017). The standard trust signal is model confidence. Our contribution is to show that for frontier-model lean classification a *free* signal—whether two independent models agree—matches calibrated confidence as a global error ranker and beats it in the operationally relevant low-budget regime, and requires no labels to compute.

**Political bias in LLMs, and news tasks.** A large literature measures LLM political orientation, largely via questionnaires (Feng et al., 2023; Santurkar et al., 2023), whose validity as a behavioral measure is contested (Röttger et al., 2024). Closer to our task, concurrent work reports “ideological center-collapse” in frontier-model news classification and steered generation on AllSides-labeled articles (Kennedy et al., 2026). We are not measuring bias but the *reliability* of a bias detector’s outputs; where we do report a directional error (§6), our per-class decomposition refines the center-collapse observation into a right-to-left asymmetry and replicates it across four labs.

**Bias detection in NLP.** Span-level media-bias detection is an established task (Spinde et al., 2021; Hamborg et al., 2019) using outlet-anchored corpora such as AllSides (Baly et al., 2020); we adopt that lineage and ground truth and study what a frontier model’s outputs on it can and cannot support.

## 3 SETUP

We reuse the corpus, models, and rollouts of the companion study; we summarize here and defer the full protocol (including the pre-committed reliability gates behind the judge instruments) to it. The corpus is  $N=200$  contemporary U.S. news articles on a balanced  $5 \times 8$  grid: five AllSides-anchored lean classes (Left, Lean Left, Center, Lean Right, Right; 40 each) crossed with eight topics, 400–1500 words, each also carrying a continuous expert lean rating. Models receive *article text only*—no outlet, headline, URL, or date—so classification cannot reduce to outlet priors.

Two frontier *target* models (Claude Sonnet 4.5, GPT-4.1) and two cross-family *judge* models (Claude Sonnet 4.6, GPT-5) perform three tasks: span-level bias detection with a shared bias-type schema (Spinde et al., 2021; Hamborg et al., 2019), summarization with represented-perspective counts, and five-class lean classification with a self-reported confidence. Two further open-weight judges (Qwen3-235B-A22B, Alibaba; DeepSeek-V3.2), added post hoc via HuggingFace Inference Providers, extend the classification committee to six models across four labs for the four-family

| Reliably agreed ( $\kappa$ ) |      | Barely above chance ( $\kappa$ ) |      |
|------------------------------|------|----------------------------------|------|
| opinion stated as fact       | 0.45 | framing                          | 0.22 |
| negativity                   | 0.32 | omission of attribution          | 0.19 |
| word choice                  | 0.31 | bias by omission                 | 0.14 |
| sensationalism               | 0.24 | unsubstantiated claims           | 0.09 |

Table 1: Cross-model agreement on bias forms (Cohen’s  $\kappa$ , two target models, 256 shared articles). Detectability tracks textual explicitness: models agree on loaded *words*, not on what a story *left out* or whether a claim is *unsupported*.

checks (`analysis/rate_articles_ext.py`). All analyses are exploratory (not pre-registered) and use expert AllSides labels as ground truth (Baly et al., 2020); agreement is Cohen’s  $\kappa$ , and bootstrap CIs use  $10^4$  resamples with a fixed seed.

#### 4 WHICH PRIMITIVES REPRODUCE

**Detection volume is a valid intensity meter.** The simplest signal—how many spans a model flags—tracks how loaded an article is: flagged-span count rises monotonically with the article’s expert-rated intensity (Spearman  $\rho=0.64$  Sonnet,  $0.58$  GPT-4.1 against  $|\text{lean rating}|$ , both  $p<10^{-18}$ ; partisan articles draw  $2.3\times$  the flags of Center articles). A builder can trust a flag *count* as a meter even where the individual flags are unreliable.

**Lean predictions track the continuous rating, not just the bucket.** Each model’s five-class prediction rank-correlates with the *continuous* expert rating at least as strongly as with the discrete label (Spearman  $0.77\text{--}0.86$  across the four models; difference vs the label  $\leq 0.01$ ), and within a single label class the models resolve graded intensity the bucket discards (within-Lean-Left  $\rho(\text{pred}, \text{rating})=0.40\text{--}0.50$ ,  $p \leq .01$ , all four). The lean signal is thus finer than the five-way label, and the results below are not an artifact of five-way bucketing. (We do not claim independence from the AllSides labels: the continuous rating and the discrete label partition this corpus identically.)

**Only concrete, word-anchored bias forms reproduce.** Given both target models the same bias-type schema, we measure how often they agree a form is present (Cohen’s  $\kappa$  over 256 jointly-rated articles, Table 1). No form clears moderate agreement—the ceiling is opinion-stated-as-fact at  $\kappa=0.45$ —and the forms defined by what is *absent from* or *unverified against* the text collapse toward chance. The disagreement is not random: the two models have distinct habits—Claude leads with bias-by-omission (14% of its flags), GPT-4.1 with sensationalism (15%)—so one hunts what a story omits while the other hunts its tone, which is exactly why they cannot agree omission is present. The unreliability reaches the span level: even when both flag the same article, the specific highlighted text overlaps at only Jaccard  $0.26$ . For a tool: show concrete word-level flags; treat any single model’s omission, framing, or unsupported-claim flag—and any individual highlight—as one model’s opinion, surfaced only on cross-model agreement.

**A cheap word-list is a corpus monitor, not a per-item detector.** A deterministic paired left/right lexicon corroborates aggregate directional patterns, but as a filter for a judge’s per-output verdict it reaches only precision  $0.25$ , recall  $0.30$ —so it belongs in a corpus-level regression check, never as a live per-article flag.

#### 5 A LABEL-FREE TRIAGE PROTOCOL

The map says most single-model outputs cannot be shown as fact. A deployed product cannot fix this with ground truth—it has none at inference—so it needs a trust signal computable from the model outputs alone. We show the best such signal is *model disagreement*, and benchmark it against the obvious alternative, model confidence.

**Route disagreement first.** On five-class lean, when the two target models agree they are  $71.4\%$  accurate ( $n=182$ ); when they disagree, only  $31.6\%$  ( $n=117$ ; gap  $39.8\text{pts}$ ,  $95\%$  CI  $[29.0, 50.4]$ ). As an error ranker, the free disagreement bit is *statistically indistinguishable* from a calibrated confidence score in global discrimination (AUC  $0.692$   $[\cdot639, \cdot744]$  vs  $0.697$   $[\cdot636, \cdot754]$ ; paired difference

| Review budget         | 10%  | 20%         | 30%         | 40%  | 50%  |
|-----------------------|------|-------------|-------------|------|------|
| route by disagreement | 15.5 | <b>31.1</b> | <b>46.6</b> | 61.3 | 67.7 |
| route by confidence   | 15.2 | 25.0        | 38.6        | 56.1 | 65.2 |

Table 2: Percent of all classification errors captured at each human-review budget (route least-reliable first;  $n=299$ , two target models). Disagreement dominates confidence in the practical 20–30% regime (bold,  $p<.05$ ).

–0.004,  $p=.90$ ), yet it captures more errors per unit of human-review budget in the operational regime, because disagreements are far more error-dense than low-confidence items (Table 2): at a 20–30% review budget, routing disagreement first captures +6 to +8 points more of all errors than a confidence threshold ( $p=.042$  and  $p=.030$ ). Integrated over *all* coverage levels the picture is complementary: confidence, being continuous, resolves risk within the agree and disagree groups and so has a modestly lower area under the risk–coverage curve (AURC 0.214 vs disagreement’s 0.300; random 0.436, oracle 0.103), even though the two are tied on error ranking. The two signals carry the same information but deliver it differently—disagreement is *free* and most powerful in the low-budget regime, confidence is smoother across the curve—and *combining* them (route disagreements first, break ties within each group by confidence) dominates either alone (AURC 0.198), an ordering stable across all five prompt conditions (disagreement AURC 0.27–0.35, confidence 0.19–0.22). The operational recommendation: *route model disagreements to human review first, rank within groups by confidence, and expect no ranking gain from confidence alone until the budget exceeds the disagreement rate*.

**At committee scale, trust unanimity—not a vote.** Adding the two judges, a four-model majority vote does *not* beat the best single model (0.646 vs 0.662 exact; McNemar  $p=.68$ ; bootstrap gain 95% CI  $[-.061, +.030]$ ), and a builder cannot pre-select which model will be best (bootstrap  $P$ (is best): GPT-4.1 .63, Sonnet 4.6 .35, GPT-5 .02, Sonnet 4.5 .00). Ensembling therefore buys robustness, not accuracy. Its deployable value is instead a high-precision *gate*: the four models agree unanimously on 44% of articles, and there they are 87% accurate—so unanimity auto-clears nearly half the volume at high precision, and the residual is exactly the disagreement set the router sends to review.

**Never read raw confidence as a probability.** Both models are severely overconfident on the exact five-class label (mean confidence 0.85–0.89 vs accuracy 0.53–0.63); a stated 0.9 empirically means 71% (Claude) to 77% (GPT-4.1) exact accuracy. GPT-4.1’s confidence is the better gate (Brier 0.286 vs 0.336; ECE 0.260 vs 0.324, the exact-target gap suggestive at  $p=.05$ ), and—critically—the overconfidence is almost entirely an *exact-boundary* effect: scored on lean *direction* (within one class), GPT-4.1’s confidence is essentially calibrated (ECE 0.024) and Claude’s nearly so (0.085). Confidence is a usable ordering for *direction*, never a usable probability for the exact label—rescale before thresholding.

## 6 THE DIRECTIONAL FAILURE MODE

The reproducible signals above are politically even-handed on average; the classification *errors* are not. Breaking accuracy down by true class exposes a single hard case shared by every model (Table 3): *Lean Right* is the hardest label for every model we evaluate—all six, spanning four independent labs (exact accuracy 0.07–0.46), and difficulty falls as an article’s expert lean strengthens (Spearman  $\rho = -0.37$ ,  $p<10^{-7}$ )—the models read the poles well and lose the moderate right. The misreads are *directional*: misclassified Lean-Right articles are pulled *left*, not merely blurred to center—33% are labeled Lean-Left or Left and 21% Center, while Center articles are called Lean-Left 32% of the time versus Lean-Right 4%. A “balanced feed” or lean-labeling product built on these classifications therefore *systematically under-counts right-leaning content*, consistent with concurrent reports of “center-collapse” (Kennedy et al., 2026), which our per-class decomposition refines into a right-to-left asymmetry.

**The map holds across four independent labs.** Adding two open-weight judges from two further families—Qwen3-235B-A22B (Alibaba) and DeepSeek-V3.2 (DeepSeek)—extends the committee to six models over four labs ( $n=198$  articles rated by all six; `analysis/four_family_analysis.py`). Every headline replicates. Lean Right is the hardest

| Model             | Left | Lean L | Center | <b>Lean R</b> | Right |
|-------------------|------|--------|--------|---------------|-------|
| Claude Sonnet 4.5 | .97  | .23    | .50    | <b>.07</b>    | .90   |
| GPT-4.1           | .88  | .47    | .75    | <b>.46</b>    | .72   |
| Claude Sonnet 4.6 | .92  | .78    | .55    | <b>.20</b>    | .80   |
| GPT-5             | .82  | .70    | .53    | <b>.42</b>    | .60   |
| Qwen3-235B        | .70  | .62    | .65    | <b>.35</b>    | .72   |
| DeepSeek-V3.2     | .85  | .45    | .65    | <b>.35</b>    | .68   |

Table 3: Exact classification accuracy by true lean class, six models across four labs (targets: Claude Sonnet 4.5, GPT-4.1; judges: Claude Sonnet 4.6, GPT-5, Qwen3-235B-A22B, DeepSeek-V3.2). Lean Right is the hardest label for every one. The bottom two rows are an exploratory four-family extension via HuggingFace Inference Providers (`analysis/rate_articles_ext.py`).

class for all six (Table 3); the agreement→accuracy signal holds on all fifteen model pairs, including the new cross-vendor ones (Qwen×DeepSeek agree 0.69 vs disagree 0.45; GPT-5×DeepSeek 0.75 vs 0.42); consensus stays a monotone accuracy signal, with six-model unanimity (35% of articles) at 90% accuracy; and the right-to-left misread is unchanged (34% of Lean-Right articles labeled Lean-Left or Left, pooled across the six). That the same failure and the same reliability signal appear in models from Anthropic, OpenAI, Alibaba, and DeepSeek makes an inherited-labels explanation harder to sustain and the pattern correspondingly more general.

**Both the map and the failure are prompt-invariant.** Re-running the two headline results under five different prompt scaffolds, the agreement→accuracy gap holds in all five (gap 0.28–0.36, bootstrap  $P(\text{gap}>0)=1.0$  in every condition) and Lean Right is the strict hardest class in all five (pooled accuracy 0.21–0.29). The pattern is a property of the models, not of one prompt. Two riders: the Lean-Right effect is Claude-driven (Claude is the worst model on it in 5/5 conditions, GPT-4.1 in only 3/5), and lean classification is least reliable on policy-abstract topics (elections/economy/climate) and most reliable on identity-concrete ones.

## 7 DISCUSSION

The results reduce to a short deployment policy for a news-bias product. (1) *Show decisions, filter prose, and rank flags by kind*: flag counts and word-anchored flags are trustworthy; omission/framing/unsupported-claim flags and individual highlights are one model’s opinion and should surface only on cross-model agreement. (2) *Triage by agreement, not raw confidence*: model disagreement is a free, label-free error signal that beats a confidence threshold in the realistic low-budget regime; combine the two for the best risk–coverage trade-off, and never expose raw confidence as a probability. (3) *Correct for the rightward blind spot*: expect moderate-right content to be under-labeled, and treat a “center” label on right-of-center content as a routing trigger, not a verdict. None of these require labels at inference; the only ground truth the protocol needs is at build time, to choose thresholds. This turns a measurement report into a reusable engineering recipe, and the corpus recipe and analysis code are released so the map can be re-derived on a new model generation cheaply.

## 8 LIMITATIONS

Three limits bound the claims. First, all findings are *exploratory* (not pre-registered) and inherit AllSides’ outlet-anchored labels as the sole ground truth (Baly et al., 2020); the directional error could partly reflect article-vs-outlet label drift, though uniform label noise cannot produce a one-sided right-to-left pattern, and its replication across four independent labs makes a pure-label explanation less likely. A second, independent label source (per-article human consensus, or a second rating provider) is the highest-value next test. Second, the triage router is validated on the baseline prompt condition and one model pair (though the reliability map and directional failure replicate across conditions and vendors); and the label-free signals so far require a *committee*—disagreement needs two models, the unanimity gate four. A single-model cold-start signal (self-consistency across temperature-sampled generations, benchmarked against the disagreement router’s AUC of 0.69) is a natural next step. Third, any LLM judge inside a tool’s own quality-control pipeline inherits the reliability problem we document for the models themselves.

## REPRODUCIBILITY STATEMENT

The corpus curation script and manifest, all task and judge prompts, the runner, the four-family judging scripts (`analysis/rate_articles_ext.py`, `analysis/four_family_analysis.py`), and per-call cost accounting are released; every number in the paper is reproduced by these scripts on the released rollouts. Bootstrap procedures use a fixed seed. Article text is redistributed only as required for reproduction, with source URLs and the deterministic curation script provided in lieu of bulk redistribution where licensing requires. [\[Repository and data links after de-anonymization.\]](#)

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## A FULL FOUR-FAMILY CONSENSUS LADDER

On the  $n=198$  articles rated by all six models (four labs), modal-label accuracy rises monotonically with the number of models agreeing:  $\geq 2/6$  agree covers 100% of articles at 0.66;  $\geq 3/6$ , 97% at 0.67;  $\geq 4/6$ , 81% at 0.73;  $\geq 5/6$ , 48% at 0.82; 6/6 (unanimous), 35% at 0.90. Unanimity across four independent vendors is the strongest label-free confidence signal available.